

Annex A

Computational Thinking Exercise: "Smart School Canteen Queue"

Section: _____

Score: _____

C# / Name: _____

Date: _____

Scenario

The PSHS school canteen is small and often gets crowded during lunch break. Students line up to buy food, but the process is slow because:

- Some students take too long to decide what to order.
- The cashier has to manually calculate totals and give change.
- There is no system to track which food items are running out.

Your group's task is to **decompose this problem** into smaller, manageable parts that could be solved with computational thinking (CT) Skills.

Step 1: Identify the Big Problem

Main Problem: _____

Step 2: Identify three to four Sub-Problems

Please list possible sub-problems:

1. _____

2. _____

3. _____

4. _____

Step 3: Define Computational Thinking Approaches

For each sub-problem, apply CT skills:

Sub-Problem	CT Skill	Example Solution

Step 4: Draw a flowchart or write a pseudocode for the identified sub-problem

Rubrics For Grading

Total Points: 20pts

Criteria & Levels of Performance

Criteria	Excellent (4)	Good (3)	Fair (2)	N.I. (1)
Identification of Sub-Problems	Identifies 3+ clear, relevant sub-problems that directly connect to the scenario.	Identifies 2–3 mostly relevant sub-problems.	Identifies 1–2 vague or partially relevant sub-problems.	Struggles to identify sub-problems or lists unrelated issues.

Application of CT Strategies	Correctly applies appropriate CT strategies (abstraction, decomposition, pattern recognition, algorithm design) to each sub-problem with clear reasoning.	Applies CT strategies to most sub-problems, with minor errors or limited explanation.	Applies CT strategies inconsistently, with weak or unclear reasoning.	Rarely applies CT strategies or misuses them.
Flowchart / Pseudocode X 2	Flowchart / Pseudocode is complete, logical, and easy to follow; shows clear steps and decision points.	Flowchart / Pseudocode is mostly complete and logical, with minor gaps or unclear steps.	Flowchart / Pseudo Code is partially complete, missing key steps or connections.	Flowchart / Pseudocode is incomplete, confusing, or missing entirely.
Reflection / Explanation	Provides thoughtful reflection on how decomposition helps problem-solving and identifies CT skills used with strong justification.	Provides adequate reflection with some justification of CT skills.	Provides limited reflection with weak or generic justification.	Provides minimal or no reflection.